MTGN605L	ADVANCED TRANSMISSION ELECTRON MICROSCOPY LABORATORY	1.0
MTGN656	ADVANCED ELECTRON MICROSCOPY	2.0
MTGN656L	ADVANCED ELECTRON MICROSCOPY LABORATORY	1.0
MLGN502	SOLID STATE PHYSICS	3.0
MTGN573	COMPUTATIONAL MATERIALS	3.0
MLGN515	ELECTRICAL PROPERTIES AND APPLICATIONS OF MATERIALS	3.0
MLGN583	PRINCIPLES AND APPLICATIONS OF SURFACE ANALYSIS TECHNIQUES	3.0
MLGN593	BONDING, STRUCTURE, AND CRYSTALLOGRAPHY	3.0
Applied Mathematics and Statistics Electives		
MATH506	COMPLEX ANALYSIS II	3.0
MATH510	ORDINARY DIFFERENTIAL EQUATIONS AND DYNAMICAL SYSTEMS	3.0
MATH536	ADVANCED STATISTICAL MODELING	3.0
MATH538	STOCHASTIC MODELS	3.0
MATH550	NUMERICAL SOLUTION OF PARTIAL DIFFERENTIAL EQUATIONS	3.0
MATH551	COMPUTATIONAL LINEAR ALGEBRA	3.0
Humanities, Arts, and Social Sciences Electives		
HASS523	ADVANCED SCIENCE COMMUNICATION	3.0

Mines' Combined Undergraduate/Graduate Degree Program:

Students enrolled in Mines' combined undergraduate/graduate program may double count up to 6 credits of graduate coursework to fulfill requirements of both their undergraduate and graduate degree programs. These courses must have been passed with B- or better, not be substitutes for required coursework, and meet all other university, department, and program requirements for graduate credit.

Students are advised to consult with their undergraduate and graduate advisors for appropriate courses to double count upon admission to the combined program.

Graduate Certificate Curriculum Requirements:

The certificate option consists of three of the four new courses, plus one additional elective chosen from the above list, for a total of 12 credits.

Graduate Certificate, Software Track

PHGN519	FUNDAMENTALS OF QUANTUM INFORMATION	3.0
CSCI581	QUANTUM PROGRAMMING	3.0
PHGN545	QUANTUM MANY-BODY PHYSICS	3.0
Elective		3.0
Total Semester Hrs		**12.0**

Graduate Certificate, Hardware Track

PHGN519	FUNDAMENTALS OF QUANTUM INFORMATION	3.0
PHGN535	INTERDISCIPLINARY SILICON PROCESSING LABORATORY	3.0
EENG/PHGN532	LOW TEMPERATURE MICROWAVE MEASUREMENTS FOR QUANTUM ENGINEERING	3.0
Elective		3.0
Total Semester Hrs		**12.0**

Program Director

Meenakshi Singh, Associate Professor, Physics

Department of Chemical and Biological Engineering

Matthew Crane, Assistant Professor

Department of Computer Science

Neil Dantam, Assistant Professor

Dinesh Mehta, Professor

Bo Wu, Associate Professor

Dejun Yang, Associate Professor

Department of Electrical Engineering

Peter Aaen, Professor

Department of Metallurgical and Materials Engineering

Geoff Brennecka, Professor

Brian Gorman, Professor

Department of Physics

Victoria Borish, Teaching Assistant Professor

Lincoln Carr, Professor

Patrice Genevet, Professor

Zhexuan Gong, Assistant Professor

Eliot Kapit, Associate Professor

Wouter Van De Pontseele, Assistant Professor

Frederic Sarazin, Professor

Jeff Squier, Professor

Affiliated Faculty

Matt Beard, Joint Appointment, NREL and Chemistry

Justin Johnson, NREL

Andriy Zakutayev, Joint Appointment, NREL and Metallurgical and Materials Engineering